

ELETRÔNICA DE POTÊNCIA

2ª LISTA DE EXERCÍCIOS

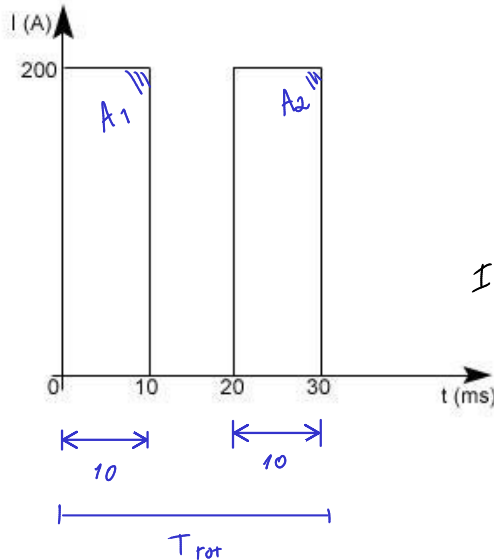
(Resolver usando o equation)

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1) Calcular o valor médio e o valor eficaz das formas de onda abaixo:

a) Def: $I_{AV} = \frac{1}{T} \int_0^T f(x) dx$



Valor médio (I_{MED}):

$$A_1 = 10 \cdot 200 \quad A_2 = 10 \cdot 200 \quad T_{TOT} = 30 \text{ ms}$$

$$= 2 \text{ k [A} \cdot \text{ms]} \quad = 2 \text{ k [A} \cdot \text{ms]}$$

$$I_{MED} = \frac{A_1 + A_2}{T_{TOT}} = \frac{2 \text{ k A} \cdot \text{ms} + 2 \text{ k A} \cdot \text{ms}}{30 \text{ ms}}$$

$$= \frac{4 \text{ k}}{30} \Rightarrow I_{MED} \approx 133,33 \text{ A}$$

$$I_{RMS} = \left| \frac{1}{T} \int_0^T f^2(x) dx \right|^{1/2} = \left| \frac{1}{T} \sum I_{PICO}^2 \Delta x \right|^{1/2}$$

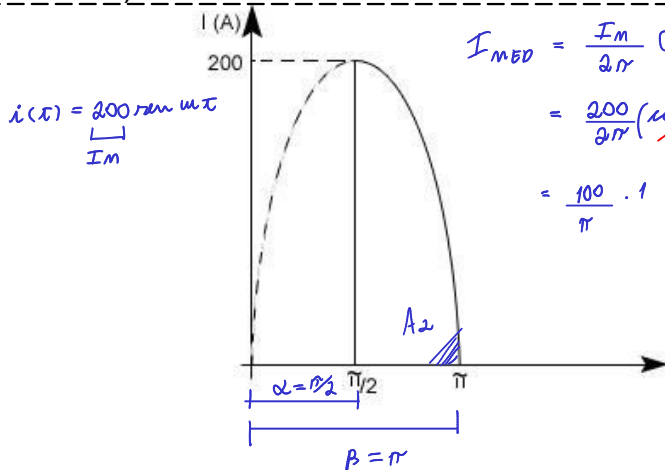
$$= \left| \frac{1}{30} [200^2 \cdot 10 + 200^2 \cdot 10] \right|^{1/2}$$

$$= \left| \frac{1}{15} \cdot 200^2 \cdot 10 \right|^{1/2} \rightarrow \left| \frac{40 \text{ k}}{1,5} \right|^{1/2}$$

$$= \left| \frac{200^2}{1,5} \right|^{1/2} \rightarrow \approx |26666,67|^{1/2} = 163,30 \text{ [A]}$$

$$I_{RMS} = 163,30 \text{ A}$$

b)



$$I_{MED} = \frac{I_M}{2\pi} (\cos \alpha - \cos \beta)$$

$$= \frac{200}{2\pi} (\cos \frac{\pi}{2} - \cos \pi)$$

$$= \frac{100}{\pi} \cdot 1 = \frac{100}{\pi} \therefore I_{MED} \approx 31,83 \text{ A}$$

$$I_{RMS} = \frac{I_M}{2} \left| \frac{\beta - \alpha}{\pi} + \frac{1}{2\pi} (\sin 2\alpha - \sin 2\beta) \right|^{1/2}$$

$$= \frac{200}{2} \left| \frac{\pi - \pi/2}{\pi} + \frac{1}{2\pi} (\sin 2\pi - \sin \pi) \right|^{1/2}$$

$$= 100 \left| \frac{1}{2} + \frac{1}{2\pi} (0 - 0) \right|^{1/2}$$

$$= 100 \sqrt{\frac{1}{2}} \approx 70,71 \text{ A}$$

$$I_{RMS} \approx 70,71 \text{ A}$$

$$I_{MED} = \frac{I_M}{2\pi} (\cos \alpha - \cos \beta)$$

$$I_{RMS} = \frac{I_M}{2} \left| \frac{\beta - \alpha}{\pi} + \frac{1}{2\pi} (\sin 2\alpha - \sin 2\beta) \right|^{1/2}$$

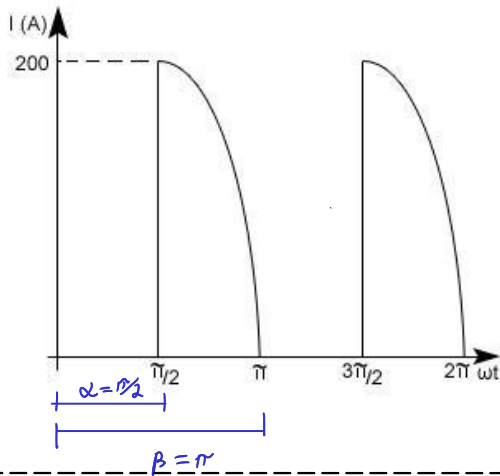
$$I_{MED} = \frac{I_m}{2\pi} (\cos \alpha - \cos \beta) \cdot 2$$

$$= \frac{200}{2\pi} (\cos \frac{\pi}{2} - \cos \pi) \cdot 2$$

$$= \frac{200}{\pi} \cdot 1 = \frac{200}{\pi} \therefore I_{MED} \approx 63,66 \text{ A}$$

$$I_{MED} = 63,66 \text{ A}$$

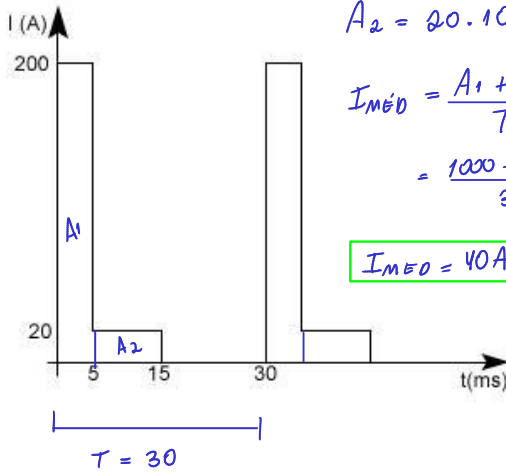
c)



$$I_{RMS} = \frac{200}{2} \left| 2 \cdot \left[\frac{\pi - \pi/2}{\pi} + \frac{1}{2\pi} (\sin 2\pi - \sin \pi) \right] \right|^{1/2}$$

$$= 100 \left| 2 \cdot \frac{1}{2} \right|^{1/2} \Rightarrow I_{RMS} = 100 \text{ A}$$

d)



$$A_1 = 200 \cdot 5 = 1000 \text{ A} \cdot \text{ms}$$

$$A_2 = 20 \cdot 10 = 200 \text{ A} \cdot \text{ms}$$

$$I_{MED} = \frac{A_1 + A_2}{T}$$

$$= \frac{1000 + 200}{30} = \frac{1200}{30}$$

$$I_{MED} = 40 \text{ A}$$

$$I_{RMS} = \left| \frac{1}{T} \sum I_{pico}^2 \Delta \tau \right|^{1/2}$$

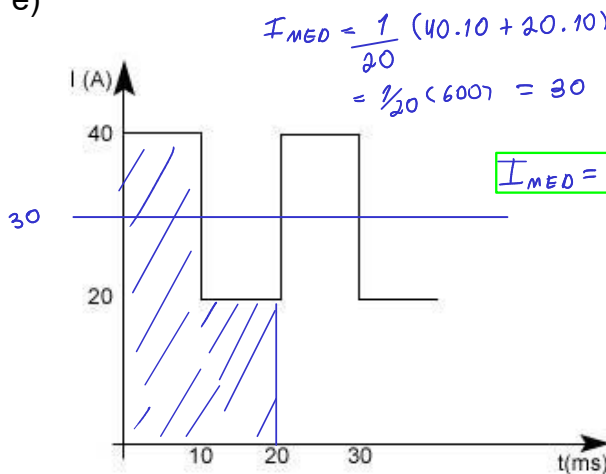
$$= \left| \frac{1}{30} [200^2 \cdot 5 + 20^2 \cdot 10] \right|^{1/2}$$

$$= \left| \frac{1}{30} [204000] \right|^{1/2}$$

$$= |6800|^{1/2} \approx 82,46 \text{ A}$$

$$I_{RMS} = 82,46 \text{ A}$$

e)



$$I_{MED} = \frac{1}{20} (40 \cdot 10 + 20 \cdot 10)$$

$$= \frac{1}{20} (600) = 30$$

$$I_{MED} = 30 \text{ A}$$

$$I_{RMS} = \left| \frac{1}{T} \sum I_{pico}^2 \Delta \tau \right|^{1/2}$$

$$= \left| \frac{1}{30} [40^2 \cdot 10 + 20^2 \cdot 10 + 40^2 \cdot 10] \right|^{1/2}$$

$$= \left| \frac{1}{30} [36000] \right|^{1/2}$$

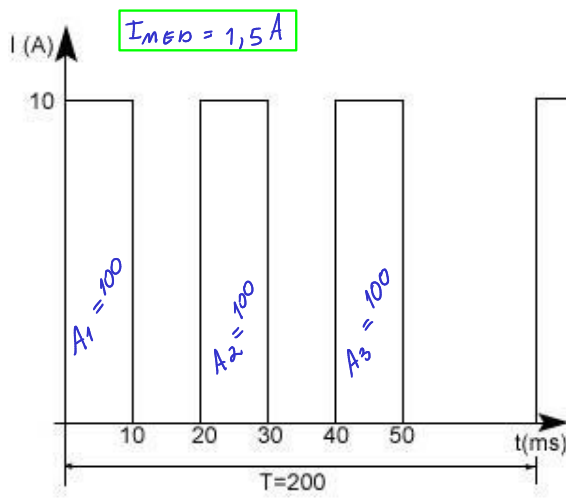
$$= |1200|^{1/2} \approx 34,64 \text{ A}$$

$$I_{RMS} = 34,64 \text{ A}$$

$$A_1 + A_2 + A_3 = 300 \text{ A.ms}$$

$$I_{MED} = \frac{1}{200} \cdot 300$$

f)



$$I_{RMS} = \left| \frac{1}{T} \sum I_{pico}^2 \Delta t \right|^{1/2}$$

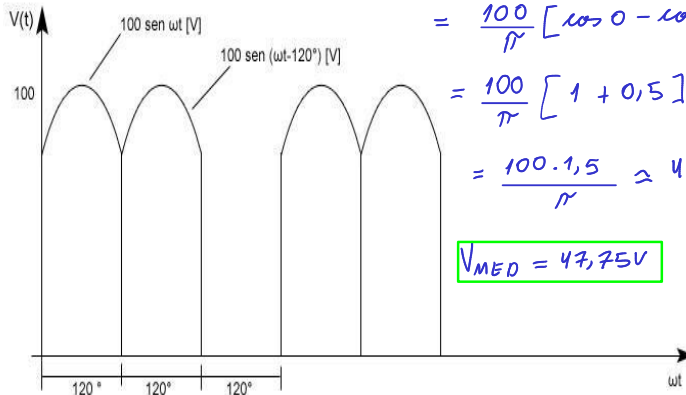
$$= \left| \frac{1}{200} [10^2 \cdot 10 + 10^2 \cdot 10 + 10^2 \cdot 10] \right|^{1/2}$$

$$= \left| \frac{1}{200} [3 \cdot 10^3] \right|^{1/2}$$

$$= \left| \frac{3 \cdot 10}{2} \right|^{1/2} = |15|^{1/2} \approx 3,87 \text{ A}$$

$$I_{RMS} = 3,87 \text{ A}$$

g)



$$\alpha = 0^\circ \quad \beta = 120^\circ$$

$$V_{MED} = 2 \cdot \left(\frac{V_m}{2\pi} [\cos \alpha - \cos \beta] \right)$$

$$= \frac{100}{\pi} [\cos 0 - \cos 120^\circ]$$

$$= \frac{100}{\pi} [1 + 0,5]$$

$$= \frac{100 \cdot 1,5}{\pi} \approx 47,75 \text{ V}$$

$$V_{MED} = 47,75 \text{ V}$$

$$V_{RMS} = \frac{V_m}{2} \left| 2 \left[\frac{\beta - \alpha}{\pi} + \frac{1}{2\pi} \sin 2\alpha - \sin 2\beta \right] \right|^{1/2}$$

$$= \frac{100}{2} \left| 2 \left[\frac{120^\circ}{\pi} + \frac{1}{2\pi} (0 + \frac{\sqrt{3}}{2}) \right] \right|^{1/2}$$

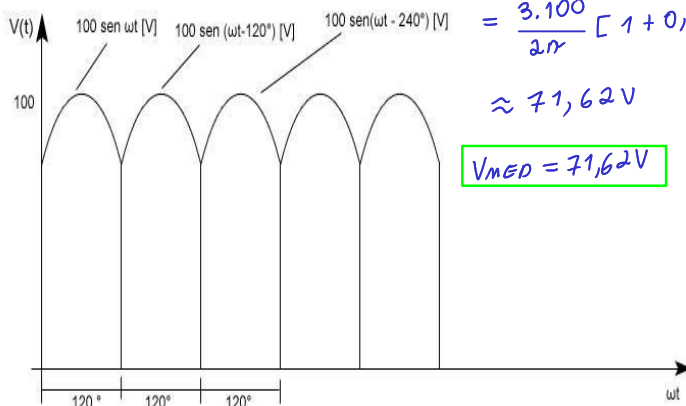
$$= 50 \left| \frac{2}{3} + \frac{\sqrt{3}}{4\pi} \right|^{1/2}$$

$$= 50 \left| \frac{4}{3} + \frac{\sqrt{3}}{2\pi} \right|^{1/2}$$

$$\approx 63,42 \text{ V}$$

$$V_{RMS} = 63,42 \text{ V}$$

h)



$$\alpha = 0^\circ \quad \beta = 120^\circ$$

$$V_{MED} = 3 \cdot \left(\frac{V_m}{2\pi} [\cos \alpha - \cos \beta] \right)$$

$$= \frac{3 \cdot 100}{2\pi} [1 + 0,5]$$

$$\approx 71,62 \text{ V}$$

$$V_{MED} = 71,62 \text{ V}$$

$$V_{RMS} = \frac{V_m}{2} \left| 3 \left[\frac{\beta - \alpha}{\pi} + \frac{1}{2\pi} \sin 2\alpha - \sin 2\beta \right] \right|^{1/2}$$

$$= 50 \left| \frac{2}{3} + \frac{\sqrt{3}}{4\pi} \right|^{1/2}$$

$$= 50 \left| 2 + \frac{3\sqrt{3}}{4\pi} \right|^{1/2} \approx 77,68 \text{ V}$$

$$V_{RMS} = 77,68 \text{ V}$$

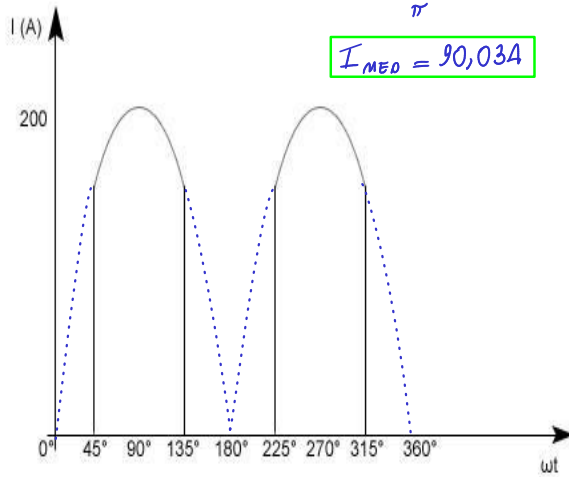
$$I_{ME0} = \frac{200}{\sqrt{2}} [\cos 45^\circ - \cos 135^\circ]$$

$$= \frac{200}{\sqrt{2}} \left[\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right]$$

$$= \frac{\sqrt{2} \cdot 200}{\sqrt{2}} \approx 90,03 \text{ A}$$

$$I_{ME0} = 90,03 \text{ A}$$

i)



$$\alpha = 45^\circ \quad \beta = 135^\circ$$

$$\frac{\pi}{4} \quad \frac{3\pi}{4}$$

$$V_{RMS} = \frac{V_m}{\sqrt{2}} \left| 2 \left[\frac{\beta - \alpha}{\pi} + \frac{1}{2\pi} \sin 2\alpha - \sin 2\beta \right] \right|^{1/2}$$

$$= 50 \left| 2 \left[\frac{3\pi/4 - \pi/4}{\pi} + \frac{1}{2\pi} \sin \frac{\pi}{2} - \sin \frac{3\pi}{2} \right] \right|^{1/2}$$

$$= 50 \left| 2 \left[\frac{1}{2} + \frac{1}{2\pi} (1 + 1) \right] \right|^{1/2}$$

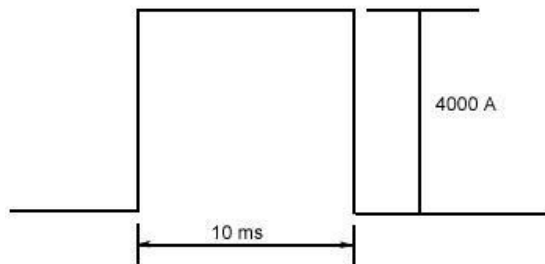
$$= 50 \left| 2 \left[\frac{1}{2} + \frac{1}{\pi} \right] \right|^{1/2}$$

$$= 50 \left| 1 + \frac{2}{\pi} \right|^{1/2} \approx 63,97 \text{ A}$$

$$I_{RMS} = 63,97 \text{ A}$$

2) Ache a capacidade I^2t das ondas abaixo:

a)



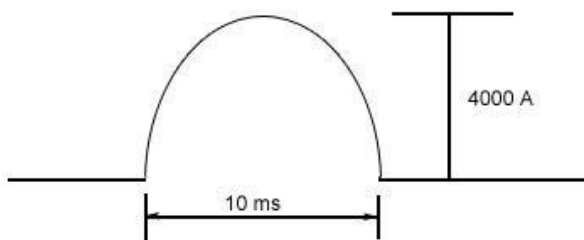
$$\sim I^2 t = I_p^2 t_p$$

$$= 10 \cdot 4000^2$$

$$= 160000 \text{ A}^2 \text{ s}$$

$$I^2 t = 160 \text{ kA}^2 \text{ s}$$

b)

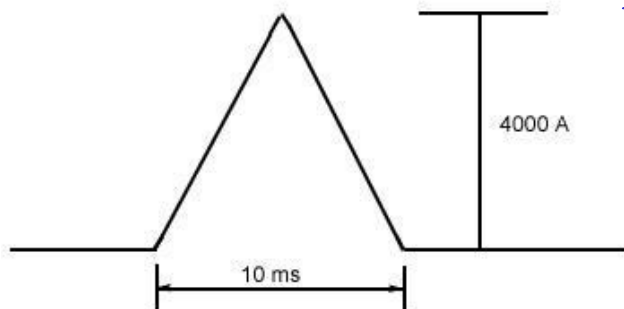


$$I^2 t = \frac{1}{2} I_m^2 t_m$$

$$= \frac{1}{2} 4000^2 \cdot 10 \text{ ms} = 80 \text{ kA}^2 \text{ s}$$

$$I^2 t = 80 \text{ kA}^2 \text{ s}$$

c)



$$I^2 t = \frac{1}{3} I_p^2 t_p$$

$$= \frac{1}{3} 4000^2 \cdot 10 \text{ ms} \approx 53 \text{ kA}^2 \text{ s}$$

$$I^2 t = 53 \text{ kA}^2 \text{ s}$$

3) a) Ache a relação da corrente de pico de uma onda triangular para retangular, com mesmo I^2t e T_p .

$$\begin{aligned}
 I^2 t_1 &\sim \frac{1}{3} I_m^2 T_p & I^2 t_2 &\sim I_p T_p & \text{se } (I^2 t)_1 &= (I^2 t)_2 \\
 \frac{1}{3} I_T^2 T_p &= I_R^2 T_p & & & \\
 I_T^2 &= 3 I_R^2 & & & \\
 I_T &= \sqrt{3} I_R & & & \\
 &= \sqrt{3} I_R & & &
 \end{aligned}$$

$$\begin{aligned}
 I_R &= \frac{I_T}{\sqrt{3}} \\
 \text{ou} \\
 I_T &= \sqrt{3} I_R
 \end{aligned}$$

b) Ache a relação dos t de uma forma de onda retangular para o de uma forma triangular, com mesmo I_p e $I^2 t$.

$$I_p^2 t_{p,R} = \frac{1}{3} I_p^2 t_{p,T}$$

$$t_{p,R} = \frac{t_{p,T}}{3} \quad \text{ou} \quad t_{p,T} = 3 t_{p,R}$$

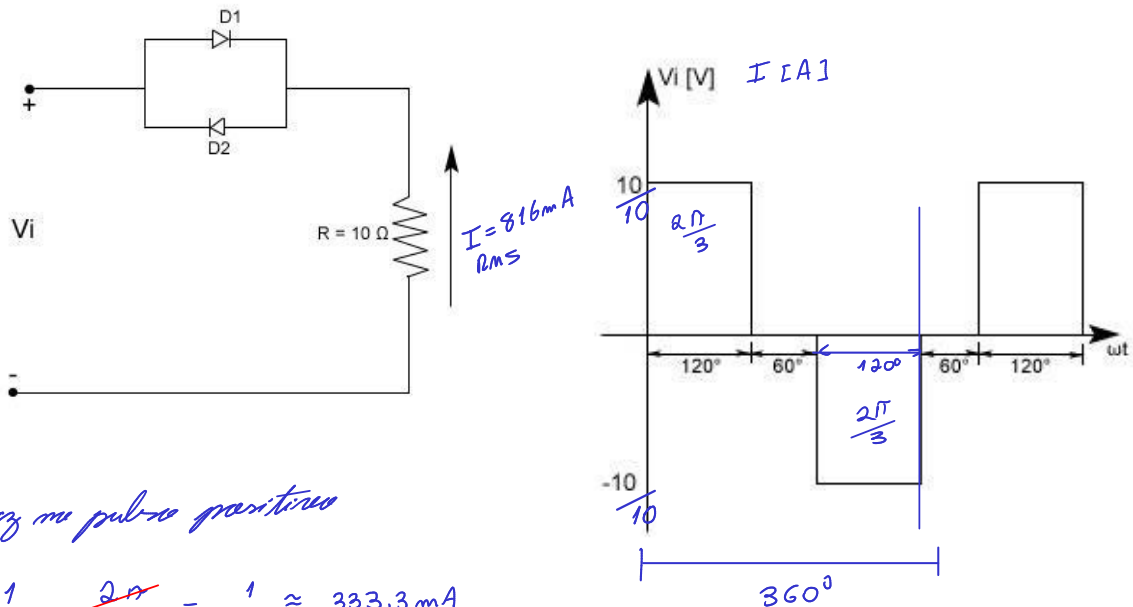
Um fusível possui um I^2t de 50000 A²s. O que acontece quando as formas de ondas do problema 2 fluem por esse fusível.

Comentar o que entende por potência média dissipada.

Para $I^2 t = 50000 \text{ A}^2 \text{s}$, as formas de onda (a), (b) e (c) possuem valores maiores do que o valor do fusível, o que resultará em seu rompimento. Assim, potência média dissipada ($P_{\text{méd}}$) é a taxa média de conversão de energia elétrica em calor por efeito Joule ao longo de um período T .

4) Calcular o valor médio e eficaz da corrente na carga e em cada diodo (para um ciclo)

considerando o circuito abaixo:



D_1 : no' conduz no pulso positivo

$$I_{MED D1} = \frac{1}{2\pi} \cdot \frac{2\pi}{3} = \frac{1}{3} \approx 333,3 \text{ mA}$$

$$I_{RMS} = \left| \frac{1}{T} \sum I_p^2 \Delta\tau \right|^{1/2} = \left| \frac{1}{2\pi} 1^2 \cdot \frac{2\pi}{3} \right|^{1/2} = \left| \frac{1}{3} \right|^{1/2} \approx 577,4 \text{ mA}$$

$$I_{MED D1} = 333,3 \text{ mA}$$

$$I_{RMS D1} = 577,4 \text{ mA}$$

D_2 : $\text{conduz no}'$ no negativo

$$I_{MED D2} = \frac{1}{2\pi} \cdot \frac{2\pi}{3} = \frac{1}{3} \approx 333,3 \text{ mA}$$

$$I_{RMS} = \left| \frac{1}{T} \sum I_p^2 \Delta\tau \right|^{1/2} = \left| \frac{1}{2\pi} 1^2 \cdot \frac{2\pi}{3} \right|^{1/2} = \left| \frac{1}{3} \right|^{1/2} \approx 577,4 \text{ mA}$$

$$I_{MED D2} = 333,3 \text{ mA}$$

$$I_{RMS D2} = 577,4 \text{ mA}$$

Para a carga:

$$V_{MED} = \frac{1}{2\pi} \left(\frac{20\pi}{3} - \frac{20\pi}{3} \right) = 0$$

$$V_{MED} = 0 \therefore I_{MED} = 0$$

$$V_{RMS} = \left| \frac{1}{T} \sum V_p^2 \Delta\tau \right|^{1/2}$$

$$= \left| \frac{1}{2\pi} \left[10^2 \cdot \frac{2\pi}{3} + (-10)^2 \cdot \frac{2\pi}{3} \right] \right|^{1/2}$$

$$= \left| \frac{1}{2\pi} \left[2 \cdot 10^2 \cdot \frac{2\pi}{3} \right] \right|^{1/2} = \left| 10^2 \cdot \frac{2}{3} \right|^{1/2} \approx 8,16 \text{ V}$$

$$S_{\text{L}} R = 10 \Omega \text{ e } V = 8,16 \Rightarrow V = R I_{RMS} \Rightarrow I = \frac{8,16}{10} \therefore I = 0,816 \text{ A}_{RMS} = 816 \text{ mA}_{RMS}$$

$$I_{MED} = 0 \text{ e } I_{RMS} = 816 \text{ mA}$$